

LINH PHUC NGO

Darmstadt, Germany / linhph.ngo@gmail.com
[LinkedIn](#) / [GitHub](#) / [Website](#) / [ORCID](#) / [Google Scholar](#)

RESEARCH PROFILE

MSc student in Artificial Intelligence and Machine Learning at TU Darmstadt and Research Assistant at Fraunhofer SIT. My research focuses on Explainable Artificial Intelligence for computer vision, with particular interests in evaluating explanation faithfulness, robustness under domain shift, and interpretable methods for object detection and dense visual prediction. I have first-author peer-reviewed publication experience and develop trustworthy AI systems for safety-critical applications in security, public health, and medical imaging.

EDUCATION

Technische Universität Darmstadt <i>M.Sc. Artificial Intelligence and Machine Learning</i>	Darmstadt, Germany <i>04.2024 – 10.2027 (expected)</i>
Unite! Research School 2025 <i>Selected Participant — AI and Cybersecurity Research Bootcamp</i>	Grenoble, France <i>11.2025</i>
Frankfurt University of Applied Sciences <i>B.Sc. Informatik</i>	Frankfurt am Main, Germany <i>10.2020 – 03.2024</i>
Vietnamese-German University <i>B.Sc. Computer Science</i>	Binh Duong, Vietnam <i>10.2019 – 12.2023</i>

PUBLICATIONS

Ngo, P. L., Pham, V. H., Bui, N. L., Phan, H. A. T., Vo, H. B., Velavan, T. P., & Tran, D. K. (2025). “**Detection of Small Water Bodies for Vector Control Using Deep Learning on Multispectral Imagery from Unmanned Aerial Vehicles.**” *Discover Artificial Intelligence*, 5. Springer Nature. DOI:10.1007/s44163-025-00422-6

RESEARCH PROJECTS

Verified Adaptive Explanations for Object Detection <i>Independent Research — First Author</i>	03.2026 – Present <i>Manuscript in Preparation</i>
- Study the reliability, faithfulness, and computational cost of post-hoc explanation methods for object detection across two detector architectures and six visual domains. - Benchmark Grad-CAM, G-CAME, D-CLOSE, LIME, and related methods using localization, perturbation-based faithfulness, saliency-map structure, and explanation-cost metrics. - Investigate metric bias toward compact Gaussian-shaped saliency maps and develop a quality-verified evaluate-then-escalate explanation pipeline.	
NEC-DINNs: Physics-Informed Forecasting of Dengue Outbreaks <i>Vietnamese-German University</i>	09.2025 – Present <i>Research Project</i>
- Develop and benchmark Neural Parameter Calibration (NPC), Disease-Informed Neural Networks (DINNs), and Neural Network Calibration-Emulation (NEC) for human-mosquito compartmental dengue models. - Evaluate forecasting accuracy, robustness, and computational efficiency using real-world dengue surveillance datasets.	
Age-Aware Face and Person Detection for Minor-Inclusive Visual Recognition <i>Fraunhofer Institute for Secure Information Technology SIT</i>	04.2026 – Present <i>Research Project</i>
- Develop computer-vision components for age-aware face and person detection within DeCNeC, a project researching AI-based approaches for ethically grounded detection of unknown harmful content involving minors. - Construct and evaluate a detection-ready dataset with paired face and body bounding boxes and fine-grained age groups to analyze performance across underrepresented minor populations. - Train and benchmark detection models to investigate the effects of dataset imbalance and annotation structure on age-aware visual recognition.	

WORK EXPERIENCE

Research Assistant (Hilfswissenschaftler), AI and Security

04.2026 – Present

Fraunhofer Institute for Secure Information Technology SIT

Darmstadt, Germany

- Conduct computer-vision research for security applications within the AI and Security team, contributing to the DeCNeC project.
- Implement and evaluate object-detection models, including YOLO and RF-DETR, through systematic experiments covering dataset construction, preprocessing, training, and benchmarking.

Research Assistant

01.2024 – Present

Vietnamese-German University

Binh Duong, Vietnam

- Conduct applied AI research at the intersection of computer vision, remote sensing, public health, and physics-informed neural networks.
- Led first-author research on deep-learning-based vector-control surveillance using multispectral UAV imagery.

AI Engineer & IT Staff (Working Student)

10.2024 – 04.2026

Fun Work GmbH

Dreieich, Germany

- Developed and deployed a production customer-service chatbot using retrieval-augmented generation (RAG) and the ChatGPT API.
- Configured and maintained cloud and on-premise infrastructure supporting production services.

Application Administration and Developer Intern

01.2023 – 05.2023

Vietnamese-German University

Binh Duong, Vietnam

- Reviewed, redesigned, and implemented a new university website using PHP and WordPress.

ACADEMIC ENGAGEMENT & SELECTED PROGRAMS

AI Grid

07.2026 – Present

Selected Member — Micro Focus Group: CV, Medical and Biological Imaging

Germany / Europe

- Selected through a competitive application process for AI Grid, a network connecting emerging AI researchers with expert communities, mentoring, and interdisciplinary collaboration.
- Participate in the Micro Focus Group on Computer Vision, Medical and Biological Imaging.

DAAD Programme: Verantwortung und Perspektive

04.2026

University Representative, Vietnamese-German University

Berlin, Germany

- Represented Vietnamese-German University at the programme “*Transnationale Bildungsangebote im internationalen Vergleich*” at the Berlin-Brandenburg Academy of Sciences and Humanities (BBAW), 13–15 April 2026.

Unite! Research School 2025

11.2025

Selected Participant — AI and Cybersecurity

Grenoble and Autrans, France

- Collaborated on a ransomware-detection use case and developed an unified AI-based framework for ransomware detection.

TECHNICAL SKILLS

Research: Explainable AI, Computer Vision, Object Detection, Robustness Evaluation, Physics-Informed Neural Networks, Remote Sensing.

Machine Learning: PyTorch, TensorFlow, OpenCV, Pandas, Matplotlib, LangChain.

Programming: Python, C/C++, R, Bash.

Tools & Platforms: Git, Docker, Jupyter, Linux, ReactJS.

Languages: Vietnamese (Native), English (C1), German (B1).